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Topic

Early detection of Cyber-Attacks on e-business platforms using artificial intelligence

Abstract

E-business platforms have become widely used in today’s digital economy, enabling seamless transactions and efficient management of business operations. These platforms offer immense convenience for businesses and consumers alike. However, they are also increasingly vulnerable to cybersecurity threats, such as SQL injection (SQLi) attacks. In SQLi attacks, hackers exploit platform vulnerabilities to gain unauthorized access to sensitive data or disrupt operations. To address these disks, deep learning algorithms can be employed to analyze data in real-time and detect potential threats early. This research report focuses on using deep learning techniques to enhance the detection and prevention of SQLi attacks on e-business platforms, improving overall security.

Introduction

Vision 2030 of the Kingdom of Saudi Arabia aims to build a diversified, knowledge-driven economy powered by innovation and technology. E-business platforms play a vital role in this transformation, enabling companies to manage operations such as customer services, digital transactions, and communication efficiently. These platforms are becoming increasingly essential for businesses by offering online solutions that reduce the need for physical infrastructure. However, as these platforms grow, they face rising risks from cyberattacks that can compromise security and erode consumer trust. Early detection of such threats is crucial for safeguarding business operations and customer data. Machine learning (ML) algorithms present a robust solution for detecting cyber threats in real time, analyzing data patterns, and flagging suspicious activities. This proactive approach helps minimize damage and secures e-business platforms, supporting the realization of Vision 2030 by ensuring the safety and reliability of digital ecosystems.

Objectives

The main objective of this work is to detect fraud in e- businesses using machine learning algorithms. To meet the needs of people objective, the following goals need to be addressed:

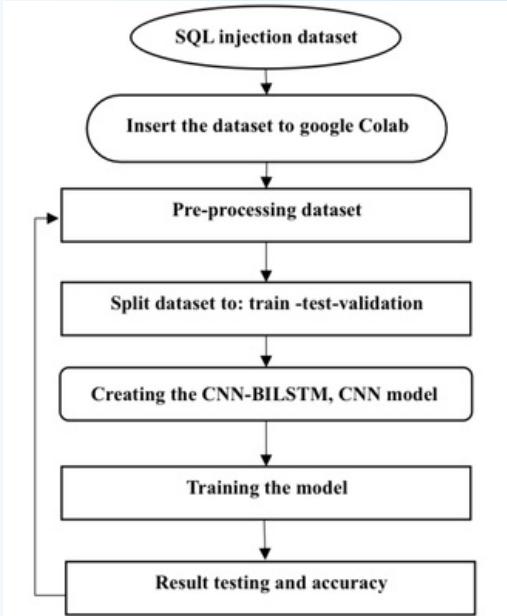
- Cyberattacks and breaches are among the most critical challenges facing information security worldwide, and early detection of these threats can help protect sensitive data and maintain the integrity of e-business platforms.
- Utilizing an artificial intelligence algorithm with high accuracy to detect and prevent cyberattacks.
- Reducing financial and operational risks for businesses and organizations.

Literature Reviews

| Name                                                                          | Problem                                                                                                                                            | Proposed Solution                                                                                                                         | Methodology                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                          | Tools                           | Result                                                                                                                                                    | Year                                                                                       |      |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------|
| SQL Injection Detection Using Machine Learning                                | SQL Injection is a major security vulnerability that allows attackers to manipulate databases, leading to unauthorized access to sensitive data.   | Detect SQL Injection attacks using machine learning algorithms. And find the best classifier                                              |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                          | Python language                 | CNN achieved 97% accuracy, which is better than other models, including Naive Bayes (95%), Logistic Regression (92%), and SVM (79%).                      | 2021                                                                                       |      |
| Cyber Threat Predictive Analytics for Improving Cyber Supply Chain Security   | The cyber supply chain faces vulnerabilities and threats that can be exploited at any point, disrupting business continuity.                       | Find the best model to predict and prevent cyberattacks in the cyber supply chain using machine learning algorithms.                      |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                          | Python language, WEKA platform, | Support Vector Machine (SVM) achieved 66% accuracy, outperforming Random Forest (62%) and Decision Tree (63%) in detecting and predicting cyber threats . | 2021                                                                                       |      |
| malware analysis and detection using machine learning algorithms              | The dataset has class imbalance, complicating malware detection.                                                                                   | Identify the best machine learning model for malware detection.                                                                           |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                          | Python Language                 | Decision Tree (DT) achieved 99% accuracy, CNN 98.76%, and SVM 96.41%, with DT performing best.                                                            | 2022                                                                                       |      |
| A Deep Learning-Based Phishing Detection System Using CNN, LSTM, and LSTM-CNN | Phishing attacks are increasingly stealing personal information by mimicking legitimate websites.                                                  | Develop a phishing detection system using deep learning to find the best model.                                                           |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                          | Python language                 | The CNN model achieved the highest accuracy of 99.2%, outperforming LSTM-CNN (97.6%) and LSTM (96.8%).                                                    | 2023                                                                                       |      |
| Evolving techniques in cyber threat hunting: A systematic review              | No unified framework for effective threat hunting. Rapidly changing attacker tactics hinder security systems' adaptability.                        | Combine traditional techniques with machine learning and AI. And Leverage threat intelligence resources for better understanding of TTPs. |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                          | Python Scikit-learn             | Random Forest (RF) 88.50% Decision Tree (DT) 95.75% Artificial Neural Network (ANN) 98.20% Naive Bayes (NB) 81.50% Logistic Regression (LR) 94.00%        | 2024                                                                                       |      |
| Name                                                                          | Problem                                                                                                                                            | Proposed Solution                                                                                                                         | Problem of CNN                                                                                                                                                                                                                                                 | Dataset                                                                                                                                                                                                                  | Methodology                     | Tools                                                                                                                                                     | Result                                                                                     | Year |
| A CNN-BiLSTM Based Approach for Detection of SQL Injection Attacks            | SQL Injection is a significant security vulnerability that allows attackers to manipulate databases and gain unauthorized access to critical data. | Detect SQL injection attacks using a CNN-BiLSTM hybrid model and compare it against other algorithms.                                     | Limited model generalization when detecting new types of SQL injection attacks -Variability in model accuracy between malicious and benign SQL queries                                                                                                         | 4,200 SQL queries labeled as normal or malicious, preprocessed for a CNN-BiLSTM model. Source: Data collected from various websites (unspecified sources).                                                               |                                 | Python and Tensor Flow.                                                                                                                                   | CNN-BiLSTM achieved 98% accuracy, outperforming Naive Bayes, Logistic Regression, and SVM. | 2021 |
| Evaluating CNN Effectiveness in SQL Injection Attack Detection                | SQL Injection attacks exploit web applications vulnerabilities, compromising database security and causing data leaks.                             | Detect SQL Injection attacks using a CNN model to differentiate malicious and benign queries, without manual feature engineering.         | -Overfitting on Training Data:The model becomes too specialized in learning from training data, making it struggle to generalize on unseen data. - Imbalanced Data: Uneven distribution of classes skews the model's predictions toward more frequent classes. | Dataset: 109,520 SQL queries labeled as malicious or benign. Sources: GitHub and Kaggle for payloads, SQLMap logs for real-world attacks, and synthetic queries created by experts, covering multiple attack techniques. |                                 | Keras and TensorFlow.                                                                                                                                     | CNN achieved 98.16% accuracy, with high precision (98.97%) and recall (97.17%).            | 2024 |

Processing (tools&Algorithms)

We compared two deep learning algorithms for detecting SQL Injection attacks in e-business platforms using Python. The first is CNN, which extracts spatial features and reduces overfitting using techniques like dropout and batch normalization. The second is a hybrid CNN-BiLSTM model, which combines CNN’s feature extraction with BiLSTM’s ability to capture sequential patterns. Both algorithms were applied following the same methodology.



Result

After using Convolutional Neural Network (CNN) and CNN-Bidirectional Long Short-Term Memory (CNN-BiLSTM) algorithms in Python to detect SQL Injection attacks, the result of using CNN was 99.29%. The CNN-BiLSTM algorithm outperformed it by achieving 99.31%.

Conclusion

With the growing threat of cyberattacks on e-business platforms, this research focused on using deep learning specifically CNN and CNN-BiLSTM for early detection of SQL injection attacks.which are among the most common and dangerous types of attacks. making their detection critical for securing online systems. Both models were implemented in Python using Keras. Results showed that CNN-BiLSTM outperformed CNN, achieving 99.31% validation accuracy compared to 99.29%. This demonstrates the strength of combining spatial and temporal feature extraction for more accurate threat detection.

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